

How to Optimise the ERNIE magnet using Compute Canada.

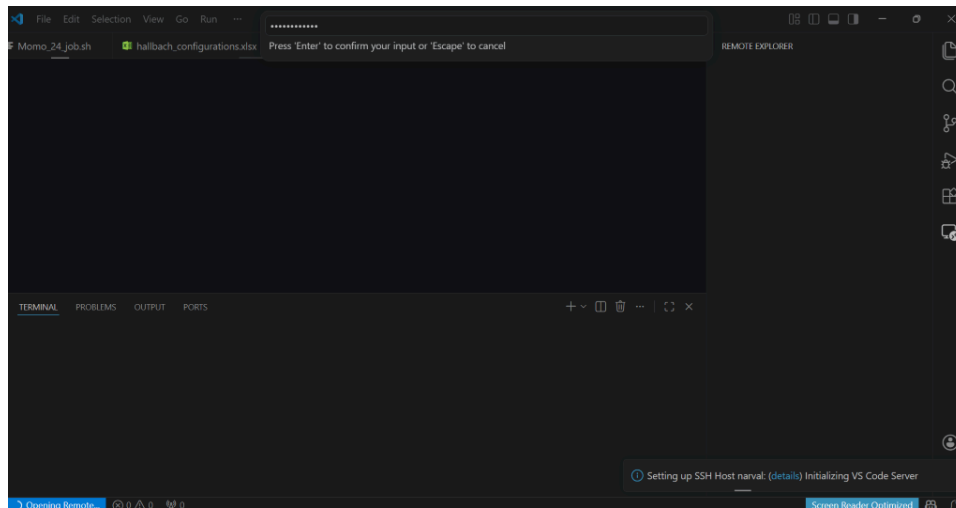
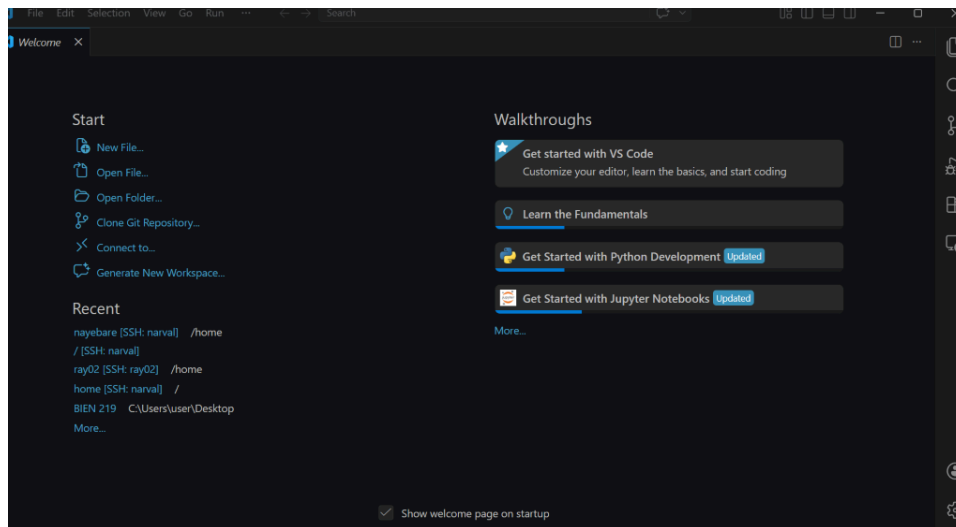
Pre-requisites: Install Visual Studio Code, the appropriate version of Python according to <https://github.com/tmachtelinckx/halbacharray-GA>

Set up a Compute Canada account.

Step 1: Open VS Code

Step 2: select ssh naval/ home

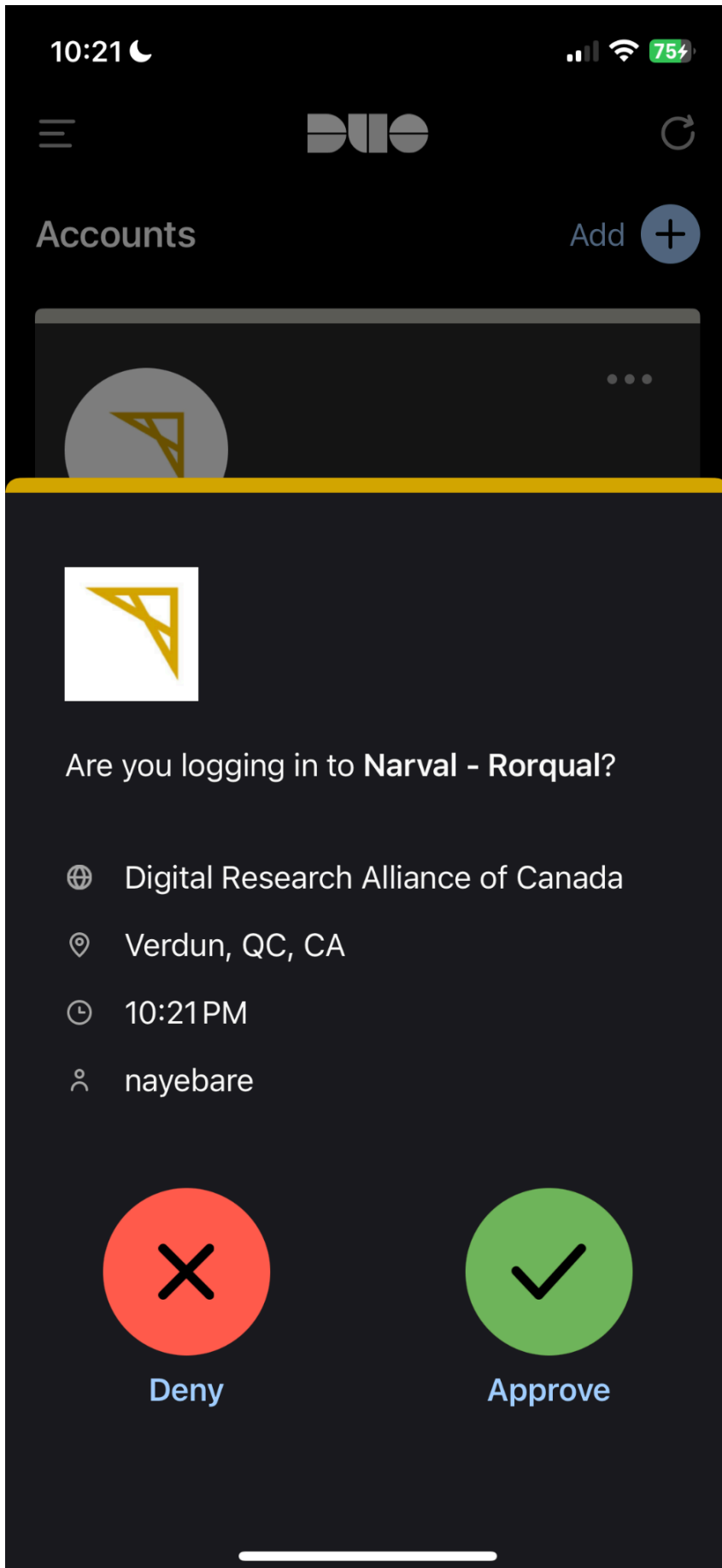
Step 3: Enter password



Step 4: Press Details

Step 5: Enter 1 and press Enter.

Step 6: Open the Duo mobile app and tap (approve).



If this does not work, do it again until it displays your personal environment on Naval.

Step 7: Then, create a new folder and name it, say, Momo_35.

Step 8: Clone the entire code from this GitHub page

(<https://github.com/tmachtelinckx/halbacharray-GA>), **config.py** files, and all into the primary folder.

Step 9: Edit the parameters in the [config.py](#) file according to the scanner preferences as shown below.

```
#-----RING DIMENSIONS-----#  
  
# Define your parameters  
InnerBoreDiameter = 80 * 1e-3 # Inner Diameter of the Ring (mm)  
OuterBoreDiameter = 160 * 1e-3 # Outer Diameter of the Ring (mm)  
magnetSize = 12 * 1e-3 # Length of cube (mm)  
  
#-----VARIABLE RING PARAMETERS-----#  
  
amountBand = np.array([1]) # Amount of bands within a ring  
bandRadiiGap = np.linspace(0, 0.1, 60) # Space between Bands (mm)  
magnetSpace = np.linspace(0, 0.05, 35) # Space Between Magnets (mm)  
bandSep = np.linspace(0.002, 0.1, 60) # Space between Bore and 1st band (mm)
```

```
ringSep = 0.024  
numRings = 7  
arrayLength = ringSep * (numRings-1)  
ringPositions = np.linspace(-arrayLength / 2, arrayLength / 2, numRings)  
  
#----- FIELD ERROR PARAMETERS -----#  
  
# Ensure that homogeneity_weight + field_strength_weight = 1  
  
T_target = 0.05 # Target field strength in Tesla  
homogeneity_weight = 0.9 # Weight for homogeneity error  
field_strength_weight = 0.1 # Weight for field strength error
```

Step 10: Edit any other necessary parameters, for example, in the [halbachFields.py](#) file, the Bremanence (bRem) value was set to 1.3 for N48 neodymium (NdFeB) magnets.

```

16 lution):
40 _mag_5)
41
42
43
44 (.025, 0.075), radius = 0.145, magnetSize = 0.0254, kValue = 2, resolution = 1000, bRem = 1.30, simDim
45
46
47
48
49
50 point=False)
51
52
53
54
55
56 mDimensions[1]*resolution)+1,int(simDimensions[2]*resolution)+1,3), dtype=np.float32)

```

Step 11: In the Momo_35 folder, create a .job file.

Step 12: In that file, paste all the job file information and edit the time for which the algorithm can run, say 24 hours.

Step 13: Then, in that job file, edit the file name to the initial file name, which is Momo_35 for this case.

```

1 #!/bin/bash
2 #SBATCH --account=def-uanazodo
3 #SBATCH --job-name=GA_Momo_35_mouse_N52_job
4 #SBATCH --nodes=1
5 #SBATCH --ntasks=24
6 #SBATCH --cpus-per-task=1
7 #SBATCH --time=24:00:00
8 #SBATCH --mem=0
9 #SBATCH --output=GA_Momo35_out.out
10 #SBATCH --error=GA_Momo35_err.err
11
12 # Load Python module (adjust version if needed)
13 module load python/3.10
14
15 # Activate virtual environment
16 source /home/nayebare/halbach_main_env/bin/activate
17
18 # Go to project directory
19 cd /home/nayebare/halbacharray-GA/halbacharray-GA
20
21 # Run script
22 python GA_main.py --results_dir /home/nayebare/halbacharray-GA/halbacharray-GA/Momo_35_mouse_N52/GA_R
23
24 #Ensure you change folder name above from Momo_35 to Momo_XX
25
26 # Deactivate environment
27 deactivate
28

```

Step 14: Press the + button at the bottom to create a terminal.

Step 15: ls, enter to see which location you are in.

Step 16: Right-click on the Momo_35 folder and copy the path.

Step 17: In the terminal, write cd space and then paste the folder path

Step 18: Then ls the components and make sure you see the Momo_35.job.sh file

Step 19: `sbatch_job` and press enter.

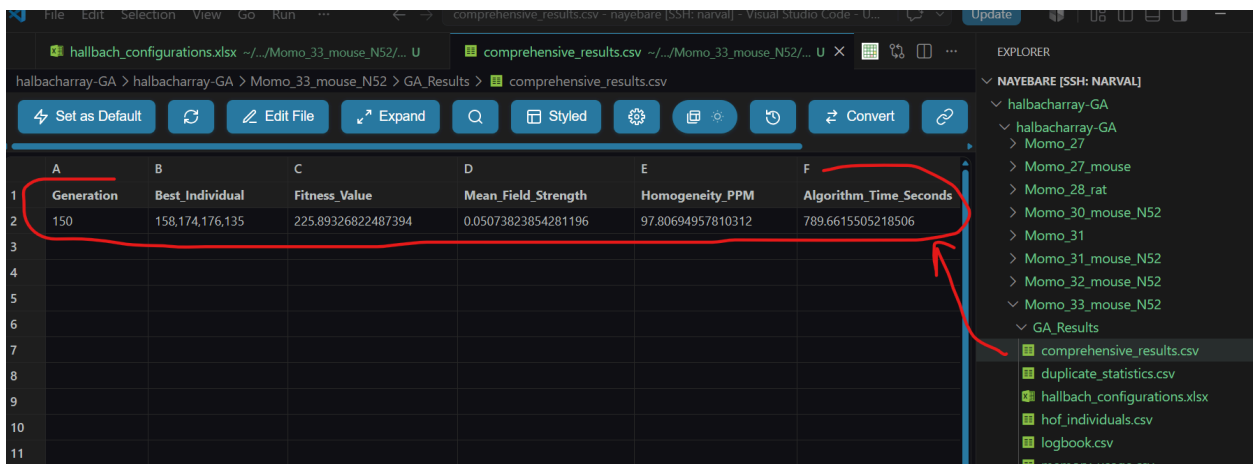
```
TERMINAL  PROBLEMS  OUTPUT  DEBUG CONSOLE  PORTS
(halbach_main_env) [nayebare@narval1 halbacharray-GA]$ ls
config.py          halbachRings.py    Momo_22_rat        Momo_28_rat        Momo_35_mouse_N52
documentation.py   initialization.py    Momo_23_rat        Momo_30_mouse_N52   Momo_36_mouse_N52
field_calculations.py Momo_17           Momo_24_mouse      Momo_31            pbs_monitor.py
GA_main.py         Momo_18           Momo_25_mouse      Momo_31_mouse_N52  __pycache__
GA_Results         Momo_19           Momo_26_mouse      Momo_32_mouse_N52
genetic_function.py Momo_20           Momo_27            Momo_33_mouse_N52
halbachFields.py   Momo_21_rat       Momo_27_mouse      Momo_34_mouse_N52
(halbach_main_env) [nayebare@narval1 halbacharray-GA]$ cd Momo_35_mouse_N52
(halbach_main_env) [nayebare@narval1 Momo_35_mouse_N52]$ ls
Momo_35_mouse_N52_job.sh
(halbach_main_env) [nayebare@narval1 Momo_35_mouse_N52]$ sbatch Momo_35_mouse_N52_job.sh
Submitted batch job 65689877
(halbach_main_env) [nayebare@narval1 Momo_35_mouse_N52]$
```

Step 20: To check if the job run, run `squeue_nayebare` and enter.

```
TERMINAL  PROBLEMS  OUTPUT  DEBUG CONSOLE  PORTS
(halbach_main_env) [nayebare@narval1 halbacharray-GA]$ ls
genetic_function.py  Momo_20           Momo_27           Momo_33_mouse_N52
halbachFields.py     Momo_21_rat       Momo_27_mouse     Momo_34_mouse_N52
(halbach_main_env) [nayebare@narval1 halbacharray-GA]$ cd Momo_35_mouse_N52
(halbach_main_env) [nayebare@narval1 Momo_35_mouse_N52]$ ls
Momo_35_mouse_N52_job.sh
(halbach_main_env) [nayebare@narval1 Momo_35_mouse_N52]$ sbatch Momo_35_mouse_N52_job.sh
Submitted batch job 65689877
(halbach_main_env) [nayebare@narval1 Momo_35_mouse_N52]$ squeue -u nayebare
JOBID    USER      ACCOUNT      NAME      ST  TIME_LEFT  NODES  CPUS  TRES_PER_N  MIN_MEM  NODELIST (
REASON)
65689877 nayebare    def-uanazodo GA_Momo_35_mou PD 1-00:00:00 1 24      N/A      0 (Priority
)
(halbach_main_env) [nayebare@narval1 Momo_35_mouse_N52]$
```

Step 21: If the job finishes, then download the relevant files.

For example, the `comprehensive_results.csv` file contains the best rings. To look at their specifications, open the `halbach_configurations.xlsx` file

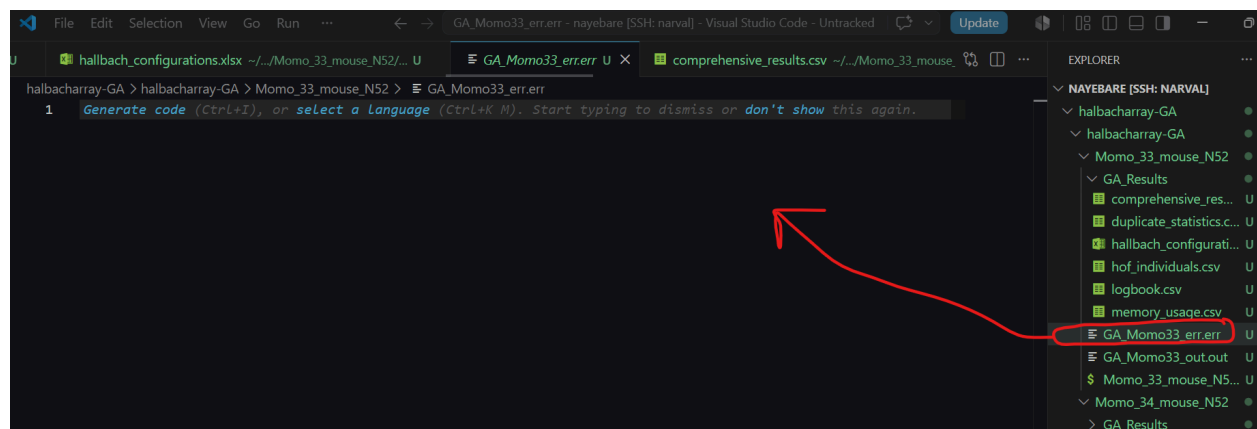


This is the output in the `halbach_configurations` file.

	A	B	C	D	E	F	G	H	I
1	Configuration_Number	Band_Number	Band_Radii_Gap_mm	Magnet_Space_mm	Band_Separation_mm	Band_1_Radius_mm	Band_1_Magnet_Count		
2		1		2.941176470588235	2	50.485	16		

Note: In case an error is incurred, check to ensure that the parameters entered in the configuration file are realistic.

The error message would be displayed here:



In that case, an output is not obtained. Edit the scanner parameters in the [config.py](#) and submit a job.